Road Pulse: Live Sentiment Analysis for UK Road Maintenance

A hackathon project turning resident frustration into actionable civic intelligence.



The Problem: £18.62 Billion Backlog

49%

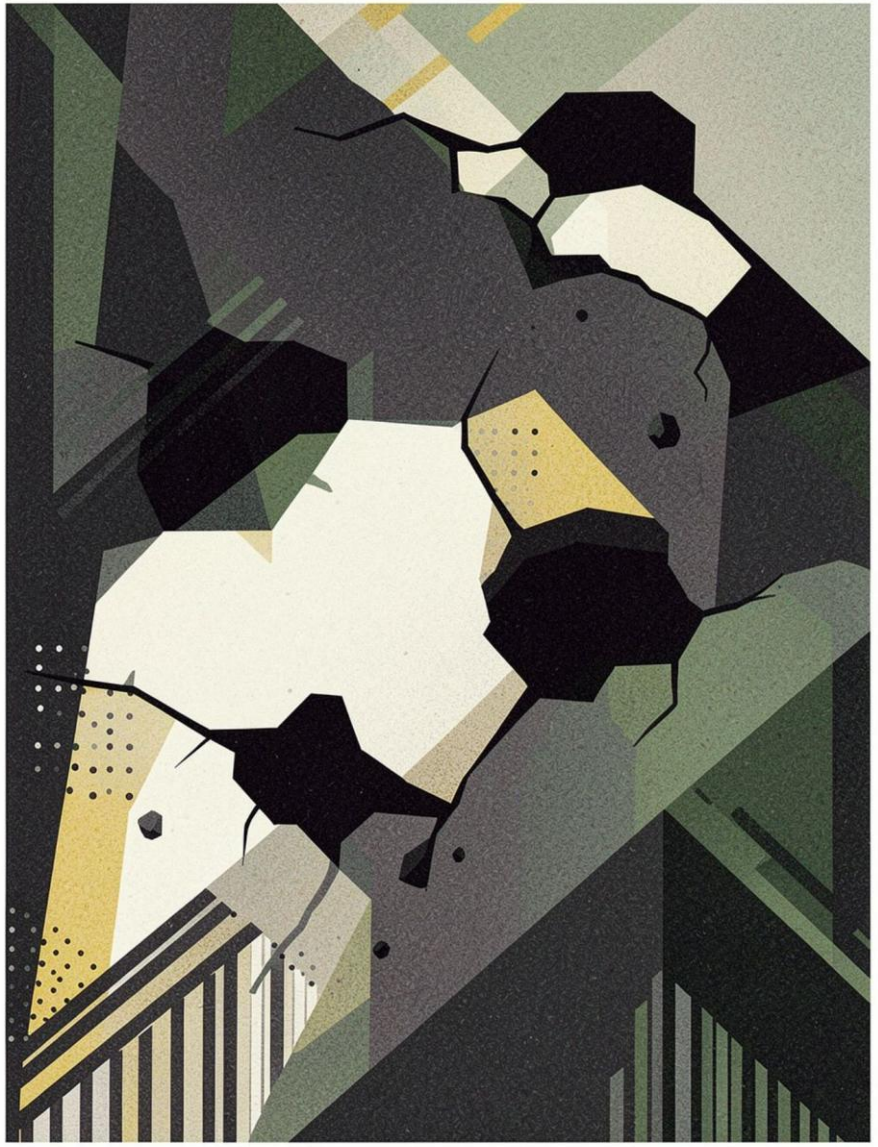
of local roads have less than 15 years of structural life remaining

+10.7%

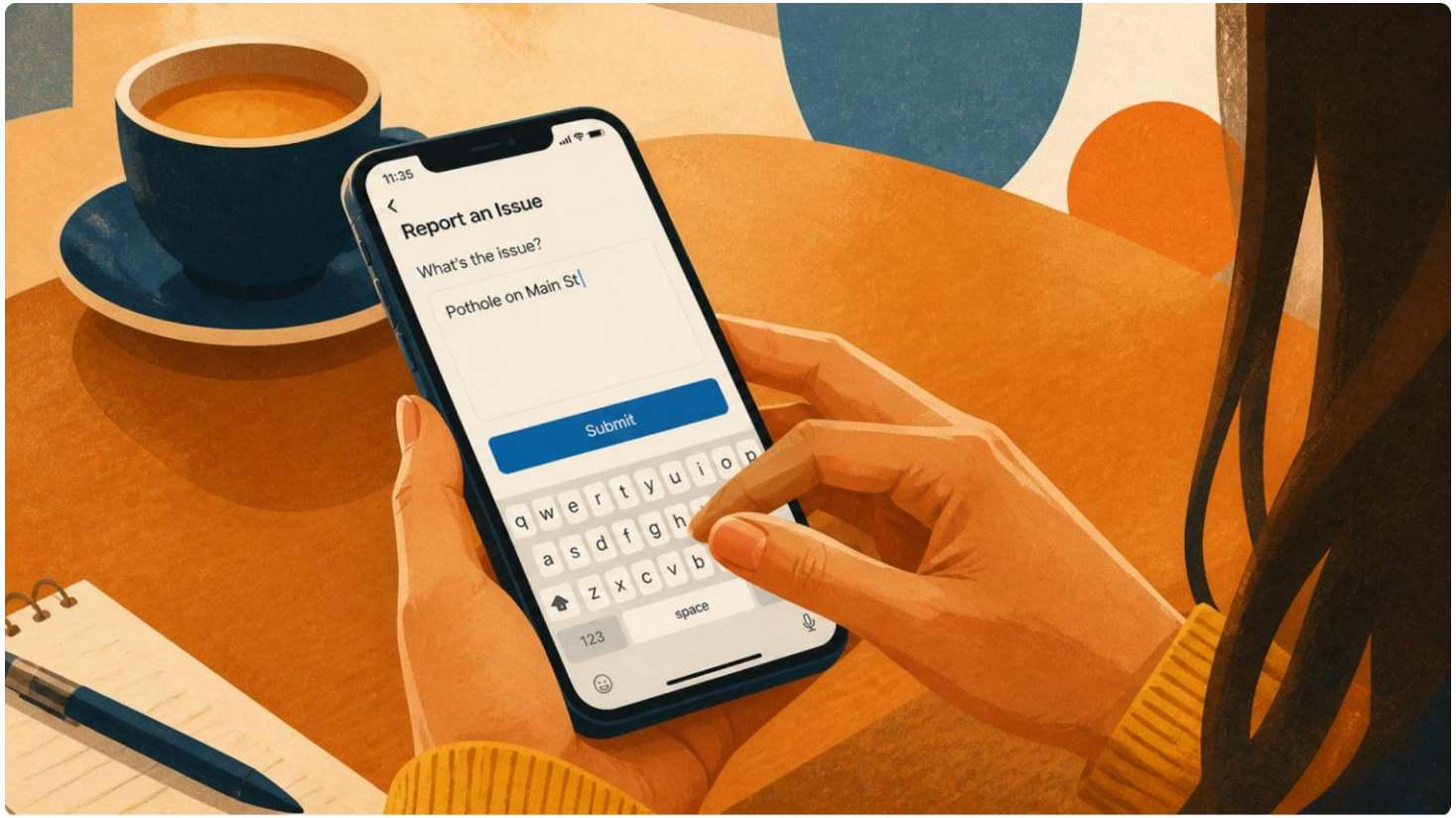
backlog growth year-over-year, per the 2026 ALARM survey

Councils

struggle to prioritise repairs and respond to growing resident frustration



The Insight: Sentiment in Reports Matters



Residents report potholes and road damage through **FixMyStreet**, but the words they use reveal far more than just facts.

→ Frustration, worry, resignation

Report language encodes emotional state, a signal hiding in plain text.

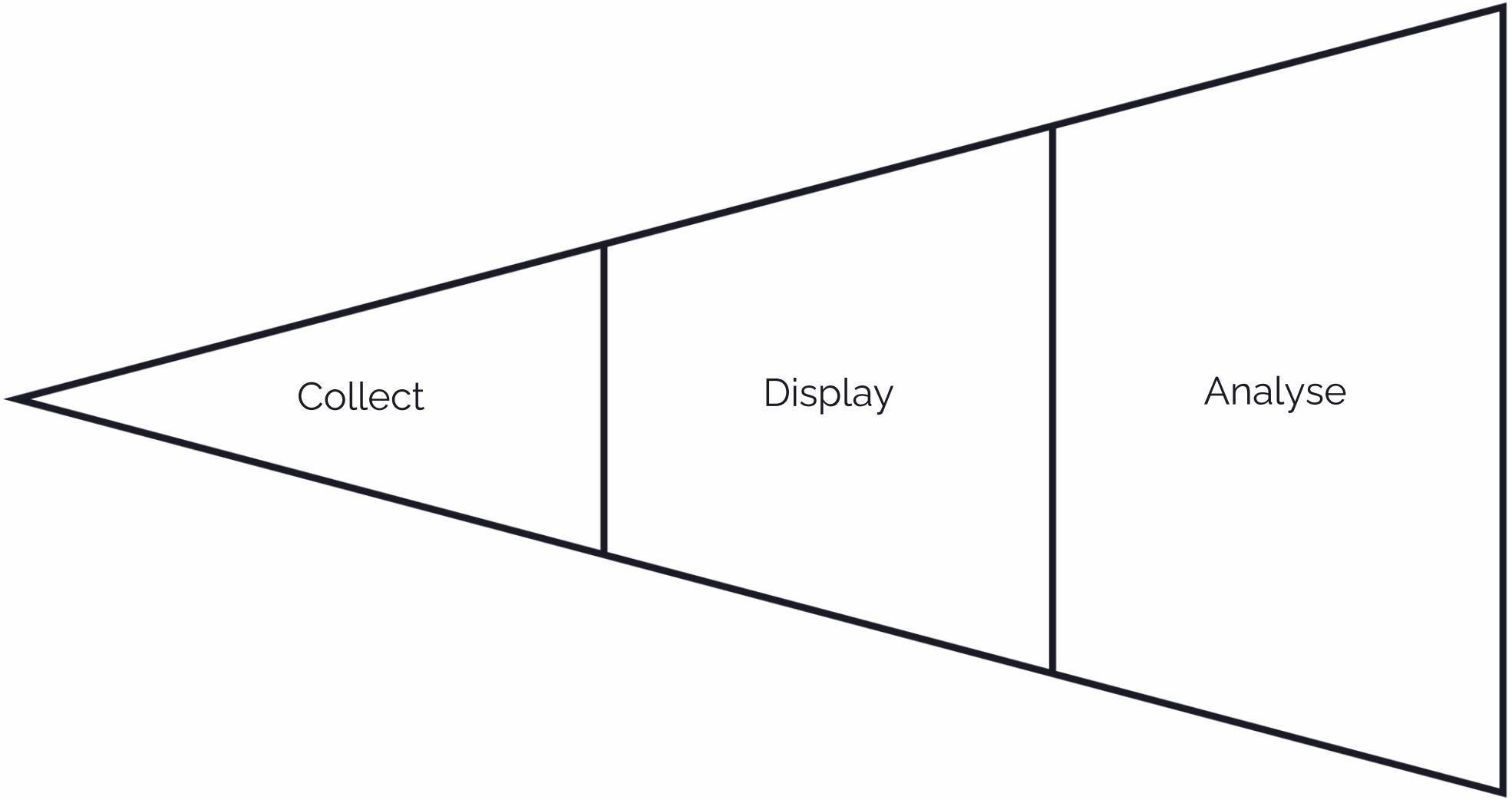
→ When sentiment shifts

Tracking changes over time shows exactly when council communication breaks down.

→ Beyond counting reports

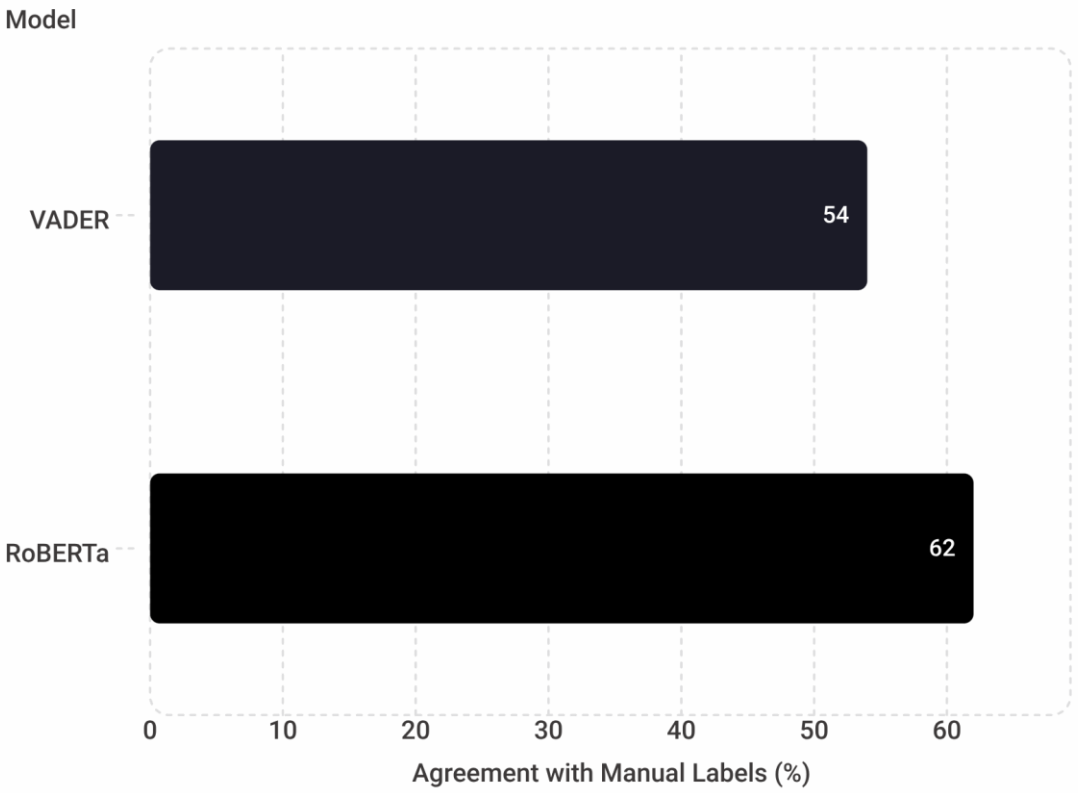
Sentiment adds a qualitative layer no simple tally can capture.

How Road Pulse Works



A fully automated pipeline from raw citizen reports to live sentiment intelligence — refreshed every 10 minutes, no manual intervention required.

The Model: RoBERTa vs. VADER



Why RoBERTa Wins

Trained on **124 million tweets**, RoBERTa understands informal, emotionally charged language. Exactly how residents write road reports.

- **62%** agreement with manual labels vs. VADER's 54%
- Correctly identified **96%** of genuinely negative reports

⚠ Known challenge: the model can flag negative *topics* (damage, hazards) as negative *sentiment* — context still matters.

What the Data Shows

1,479

Reports Collected

Across Oct 9–12, 2026

111

Councils Represented

Nationwide coverage

71%

Classified Negative

29% neutral · 0% positive

18%

Surrey County Council

Largest single contributor — highlighting stark regional variation

Ethical Guardrails Built In



Responsible AI isn't an afterthought, it's designed into every layer of Road Pulse.

Unreliable Data Warnings

Periods with fewer than 5 reports are automatically flagged as statistically unreliable.

Sentiment ≠ Road Condition

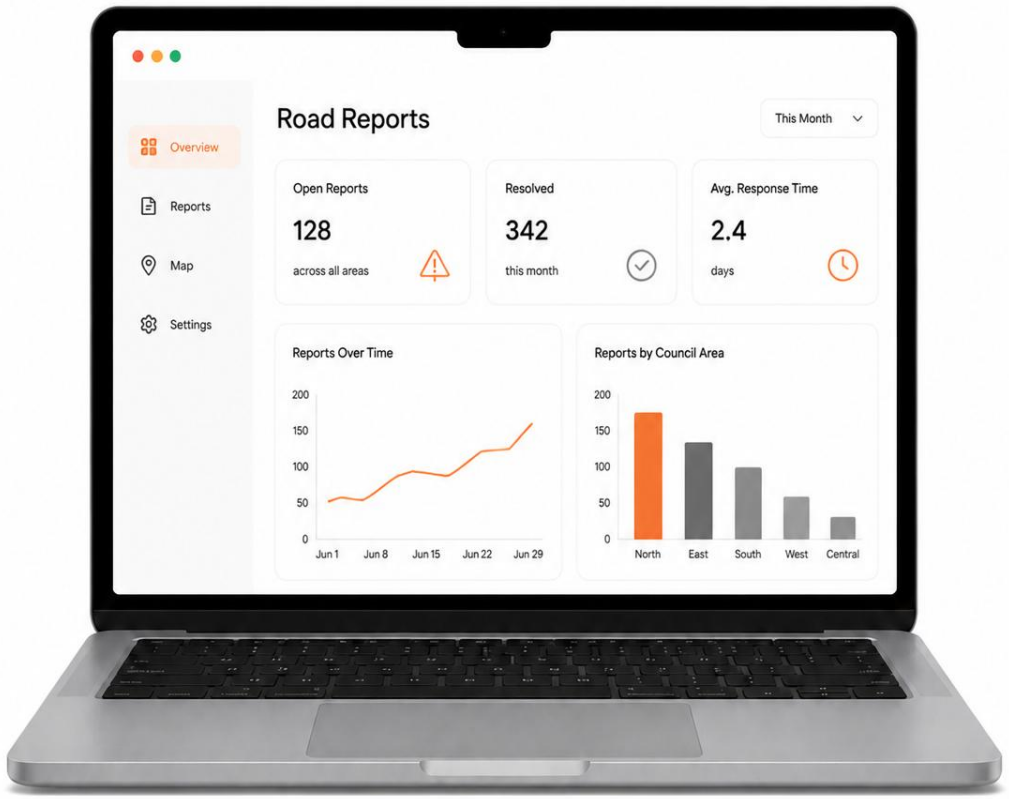
Explicit on-screen warnings prevent misreading emotional tone as a physical defect score.

Full Transparency

Users can inspect the individual reports behind any peak or dip in the trend line.

No Staff Ranking

The dashboard is not designed to judge individuals or rank council teams.



Live Dashboard & Next Steps

Try It Now

Live: road-pulse-csahn7rlvryu8wmhxjjvyb.streamlit.app

Aligned with **SDG 9:** Industry, Innovation, and Infrastructure.

What's Next

- Larger labelled dataset to fine-tune RoBERTa on road-specific language
- Automatic PII removal before any text is processed
- Expanded council coverage and longitudinal trend analysis